BM

One-stage methods

PLS1
LASSO
FPLS1
FLIN
APLS
W-PP-TMEAN
W-PP-MODE
W-PP-MEDIAN
MAP-PP-TMEAN
MAP-PP-MODE
MAP-PP-MEDIAN

0.63 0.66 0.69 0.72
RMSE

Two-stage methods

PLS+R
PCA+R
MANUAL+R
FPLS+R
FPCA+R
W-VS-TMEAN+R
W-VS-MODE+R
W-VS-MEDIAN+R
MAP-VS-TMEAN+R
MAP-VS-MODE+R
MAP-VS-MEDIAN+R

0.63 0.66 0.69 0.72
RMSE

FBM

One-stage methods

0.63 0.66 0.69 0.72
RMSE

Two-stage methods

0.63 0.66 0.69 0.72
RMSE

OU

One-stage methods

0.64 0.68 0.72
RMSE

Two-stage methods

0.64 0.68 0.72
RMSE

SE

One-stage methods

PLS1
LASSO
FPLS1
FLIN
APLS
W-PP-TMEAN
W-PP-MODE
W-PP-MEDIAN
MAP-PP-TMEAN
MAP-PP-MODE
MAP-PP-MEDIAN

0.64 0.68 0.72 0.76
RMSE

Two-stage methods

PLS+R
PCA+R
MANUAL+R
FPLS+R
FPCA+R
W-VS-TMEAN+R
W-VS-MODE+R
W-VS-MEDIAN+R
MAP-VS-TMEAN+R
MAP-VS-MODE+R
MAP-VS-MEDIAN+R

0.64 0.68 0.72 0.76
RMSE

Our RKHS methods
Reference methods
Mean of reference methods